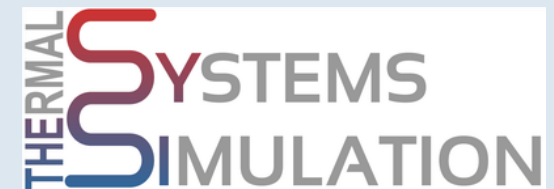


Modelica, Dymola and IDEAS

Crash Course 2026



Lucas Verleyen, Lone Meertens, Hannelore Scheipers,
Jaap Neven en Louis Hermans



Who are we?

Lucas Verleyen



Lone Meertens



Hannelore Scheipers



Jaap Neven



Louis Hermans



The SySi Team

KU LEUVEN

- Led by Professor Lieve Helsen

*To sustainably use resources through **integration and optimization of thermal systems** performance in the built environment, including other energy vectors and sectors.*



Buildings Physics Research Group



- Led by Professor Arnold Janssens

To enable sustainable, durable and energy-efficient buildings through research on building physics, indoor climate and building performance.

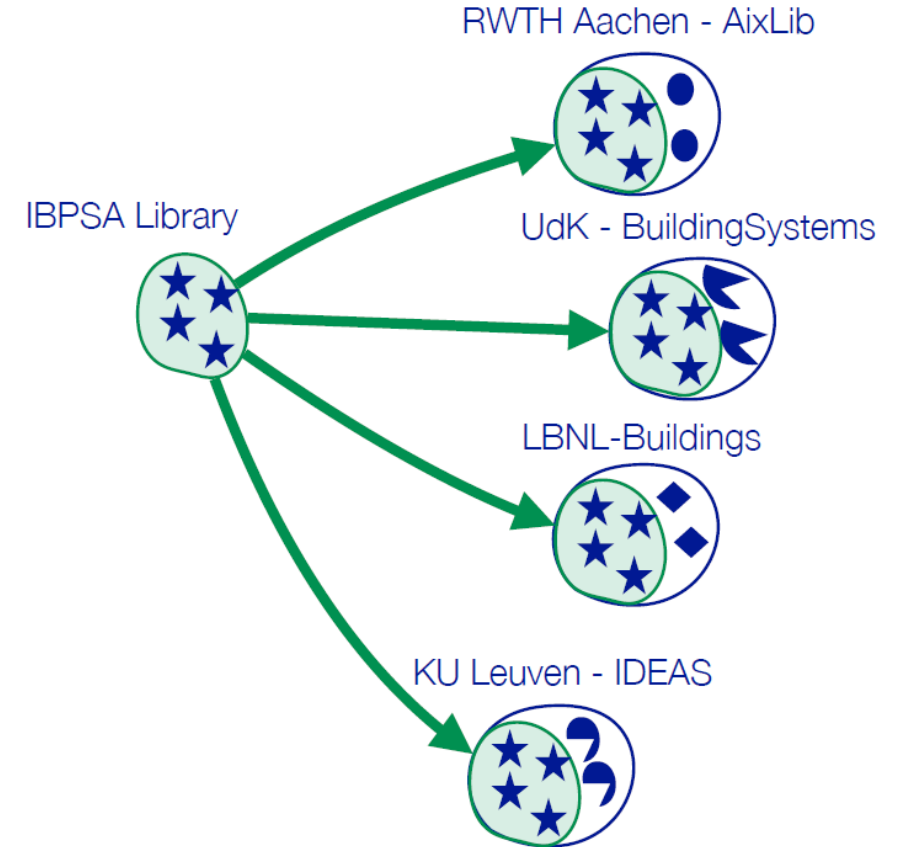


Motivation

- Why a crash course?
 - Introduction for our students and PhD students
 - Others who are interested in using Modelica
 - Broaden the user base and open-source community
- About IDEAS
 - Integrated District Energy Assessment Simulations
 - Modelica users and library development since 2010
 - IDEAS was not the only one

Motivation

- In 2013
 - Modelica for buildings was very fragmented
 - Libraries were incompatible
 - Libraries replicated each other
 - Best practices were not understood
- Joint efforts started (Annex 60, IBPSA Project 1)
- Avoid fragmentation
 - Collaborate on development
 - Implement best practices
 - Share everything open-source and free



IDEAS

- Latest release: v4.0
- BaseClasses inherited from IBPSA project 1
- Focus on buildings (thermal)
- Work on districts in other libraries (e.g. MoPED)
- Many models are validated in academic research
- Main user base: researchers, students



Agenda

Morning: Dymola and Modelica

- 9:30 - 10:00 **Lecture 1**
 - What is Modelica? What is Dymola? What is OpenModelica?
 - Modelica/Dymola basics
- 10:00 - 10:30 Exercise 1
- 10:30 - 10:45 Break
- 10:45 - 11:30 **Lecture 2**
 - Create new models/packages
 - Modelling with several components
 - Use connectors
 - Set parameters/propagate parameters
- 11:30 - 12:30 Exercise 2
- 12:30 - 13:30 Lunch break

Afternoon: IDEAS

- 13:30 - 14:00 **Lecture 3**
 - What is IDEAS?
 - IDEAS building components
 - IDEAS workflow
- 14:00 - 16:00 Exercise 3
- 16:00 - 16:30 Break
- 16:30 - 16:45 **Lecture 4**
 - IDEAS HVAC components
 - Hydronic models
- 16:45 - 18:00 Exercise 4

Part 1: Introduction to Modelica and Dymola

Lucas Verleyen

Modelica



Modelica is a **modelling language** for modelling physical systems

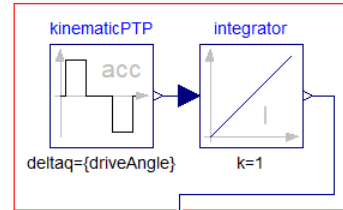
- Language specification is open source
- Object-oriented
- Acausal modelling (equation-based)
- Multi-domain
- Primarily for simulation, but usable for optimisation
- Small and large models (> 100 000 equations)
- Large community with many model libraries, especially in automotive industry (free and commercial)
- Textual and graphical modelling

Modelica

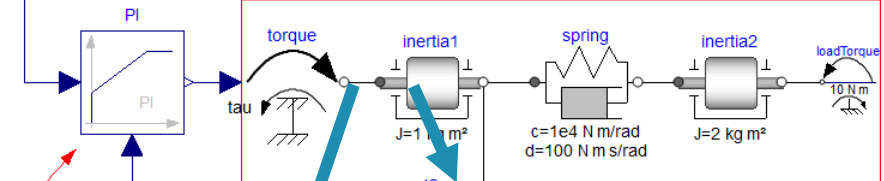
Object-oriented physical equation-based modelling

- A model represents a physical component
- Component is composed of sub-components and/or is described by equations
→ hierarchical structure
- Components can be connected to each other using connectors (=physical coupling)
- To simulate Modelica models, a Modelica simulation environment is needed

reference speed generation



PI controller



plant (simple drive train)

```
model Inertia "1D-rotational component with inertia"
  parameter SI.Inertia J(min=0, start=1) "Moment of inertia";
  parameter StateSelect stateSelect=StateSelect.default
    "Priority to use phi and w as states"
    a ;
  SI.Angle phi(stateSelect=stateSelect)
    "Absolute rotation angle of component"
    a ;
  SI.AngularVelocity w(stateSelect=stateSelect)
    "Absolute angular velocity of component (= der(phi))"
    a ;
  SI.AngularAcceleration a
    "Absolute angular acceleration of component (= der(w))"
    a ;

equation
  phi = flange_a.phi;
  phi = flange_b.phi;
  w = der(phi);
  a = der(w);
  J*a = flange_a.tau + flange_b.tau;
a
end Inertia;
```

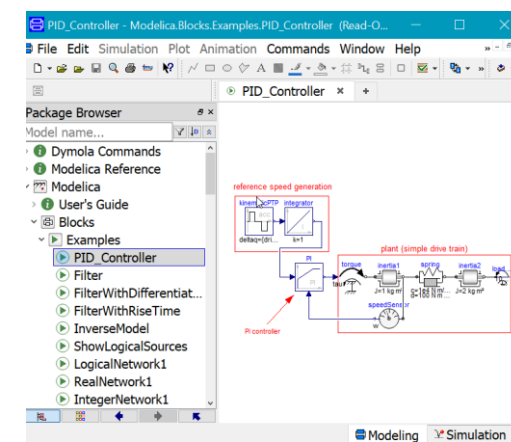
```
connector Flange_a
  "1-dim. rotational flange of a shaft (filled square icon)"
  SI.Angle phi "Absolute rotation angle of flange";
  flow SI.Torque tau "Cut torque in the flange";
a
end Flange a;
```

Dymola

Dymola is a **commercial Modelica simulation environment**

Live demo of features:

- Icon, Diagram, Editor, Info
- Package browser, modelling, simulation
- Set up (compiler), run
- Adapt parameter
- Load libraries
- Look at simulation results: plot, zoom, filter variable, plot as a function of other variable.
- Try Simulate and plot (IDEAS library)
- Open sub-components
- Documentation

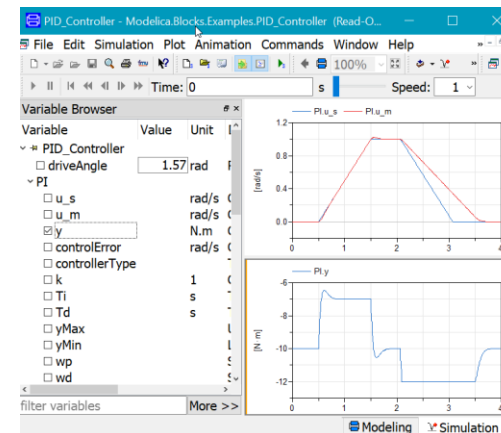


- Graphical editor
- Modelica simulation environment

```
model PID_Controller
  "Demonstrates the usage of a Continuous.LimPID controller"
  extends Modelica.Icons.Example;
  parameter Modelica.SIunits.Angle driveAngle=1.57
    "Reference distance to move";
  Modelica.Blocks.Continuous.LimPID PI (
    k=100,
    Ti=0.1,
    yMax=12,
    Ni=0.1,
    initType=Modelica.Blocks.Types.InitPID.SteadyState,
    limitsAtInit=false,
    controllerType=Modelica.Blocks.Types.SimpleController.PI,
    Td=0.1) a;
  Modelica.Mechanics.Rotational.Components.Inertia inertial(
    phi(fixed=true, start=0),
    J=1,
    a(fixed=true, start=0)) a;

  Modelica.Mechanics.Rotational.Sources.Torque torque a;
```

- Textual description (Modelica language)

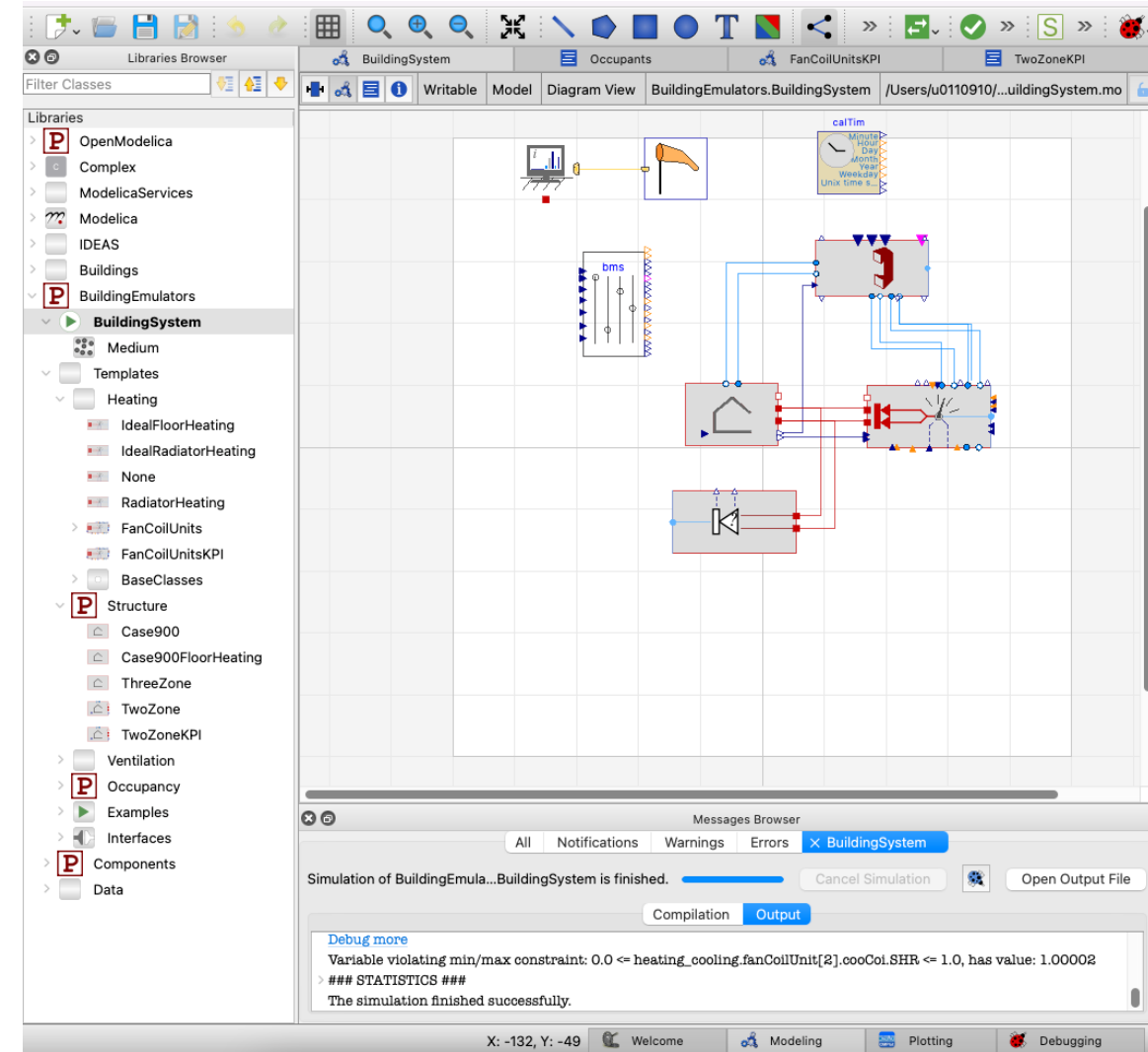
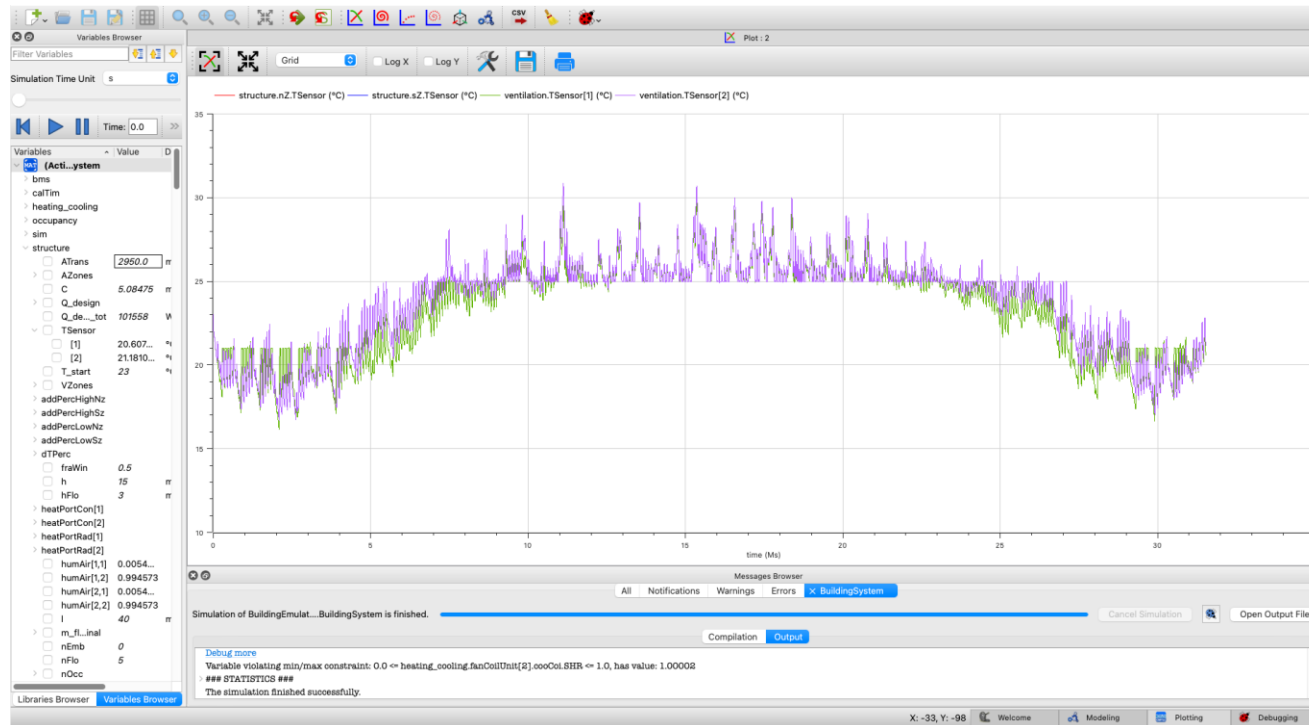


- Translation of Modelica code into executable C-code
- Coupling with a solver
- Visualization of results

OpenModelica

OM is a **free Modelica simulation environment**

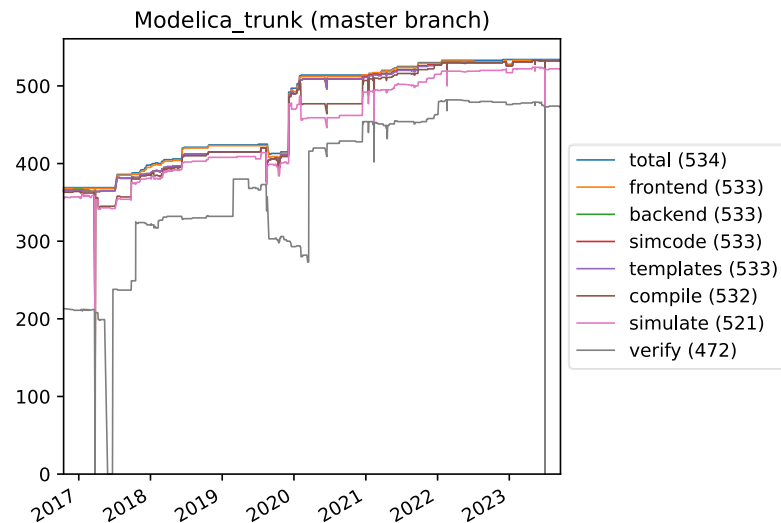
Has very similar features to Dymola, like those we have just seen



OpenModelica

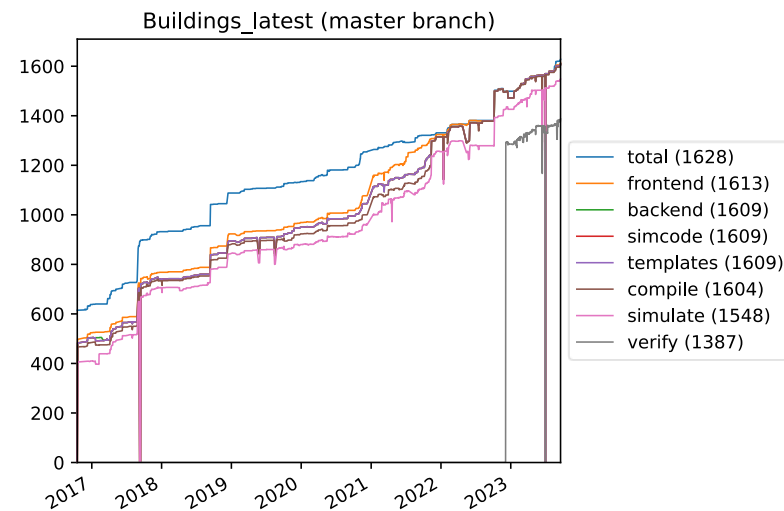
Historically less performant than Dymola, but has radically improved over the last years:

Modelica Standard Library (MSL) coverage



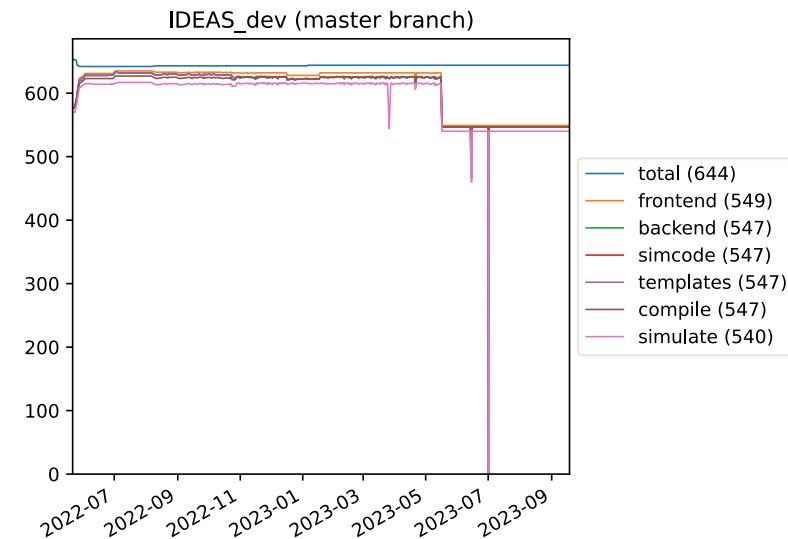
https://libraries.openmodelica.org/branches/history/master/Modelica_trunk.svg

Buildings library coverage



https://libraries.openmodelica.org/branches/history/master/Buildings_latest.svg

IDEAS coverage (recently added!)



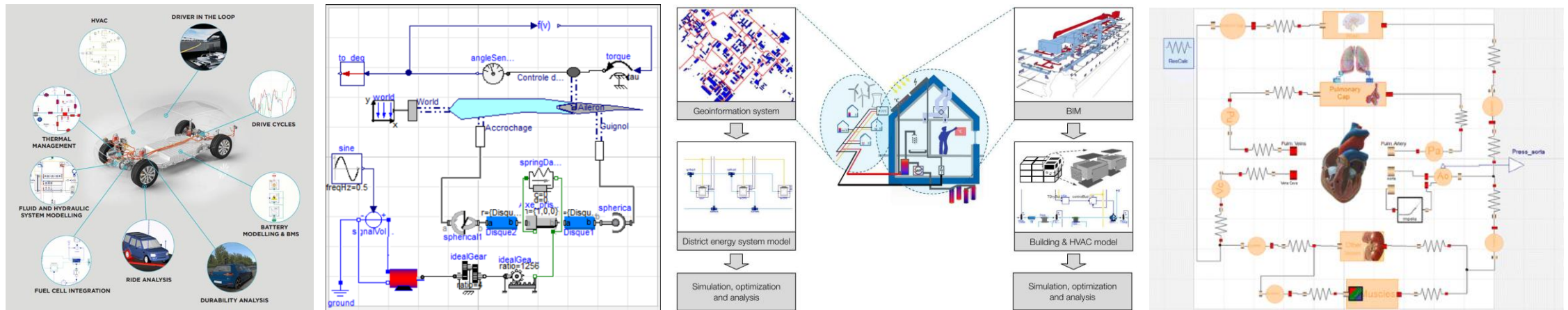
https://libraries.openmodelica.org/branches/history/master/IDEAS_dev.svg

Reasons not to use it in the crash course: Dymola is still more performant...

Applications

System modeling and simulation applications

- ➔ Automotive, aerospace, architecture, energy, medicine, ...
- ➔ In this crashcourse: thermal systems in buildings



Useful links

General

- <https://modelica.org/>
- <https://openmodelica.org/>
- <https://jmodelica.org/> **DISCONTINUED**
- <https://www.claytex.com/tech-blog/>

Modelica language

- <https://mbe.modelica.university/>
- <https://doc.modelica.org/>
- <https://specification.modelica.org/>

Libraries:

- IDEAS
<https://github.com/open-ideas>
- Buildings
<https://simulationresearch.lbl.gov/modelica>
(look at Buildings.Examples.Tutorial)
- IBPSA Project 1
<https://github.com/ibpsa/modelica-ibpsa>

Dymola user guide

- Online
- Via Dymola > help

Exercise 1

- You can find the exercise on GitHub
 - Go to www.github.com/open-ideas
 - [_CrashCourse_](#)
 - Exercises
 - Exercise 1
 - Exercise1.pdf
- https://github.com/open-ideas/_CrashCourse_/blob/master/Exercises/Exercise%201/Latex/Exercise1.pdf

Part 2: Modelling and simulating in Dymola

Lone Meertens

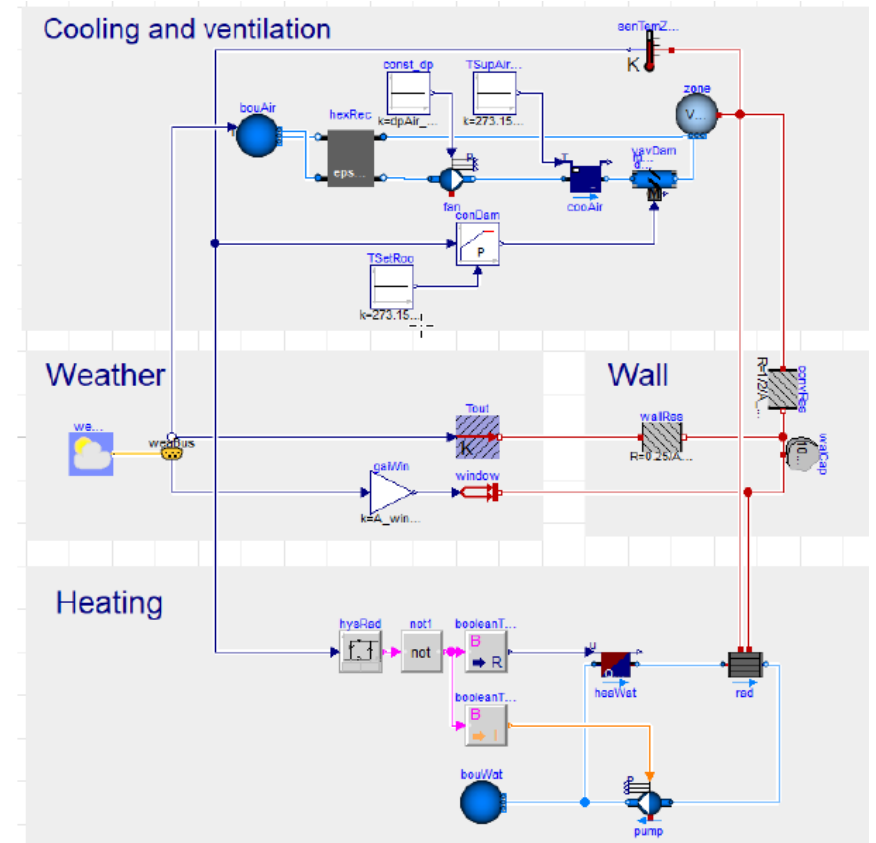
Live demonstration

- Create package, create model (+ save contents in one file)
- Search, drag and drop subcomponents. Instantiate model convention
- Connect components
- Graphical view and text view
- Simulation tab and adapt parameters
- Propagate parameters
- Units
- Use check/translate in Dymola and debug
- Advanced Modelica concepts
 - Extend
 - Replaceable

Exercise 2

- You can find the exercise on GitHub
 - Go to www.github.com/open-ideas
 - _CrashCourse_
 - Exercises
 - Exercise 2
 - Latex
 - Exercise2.pdf

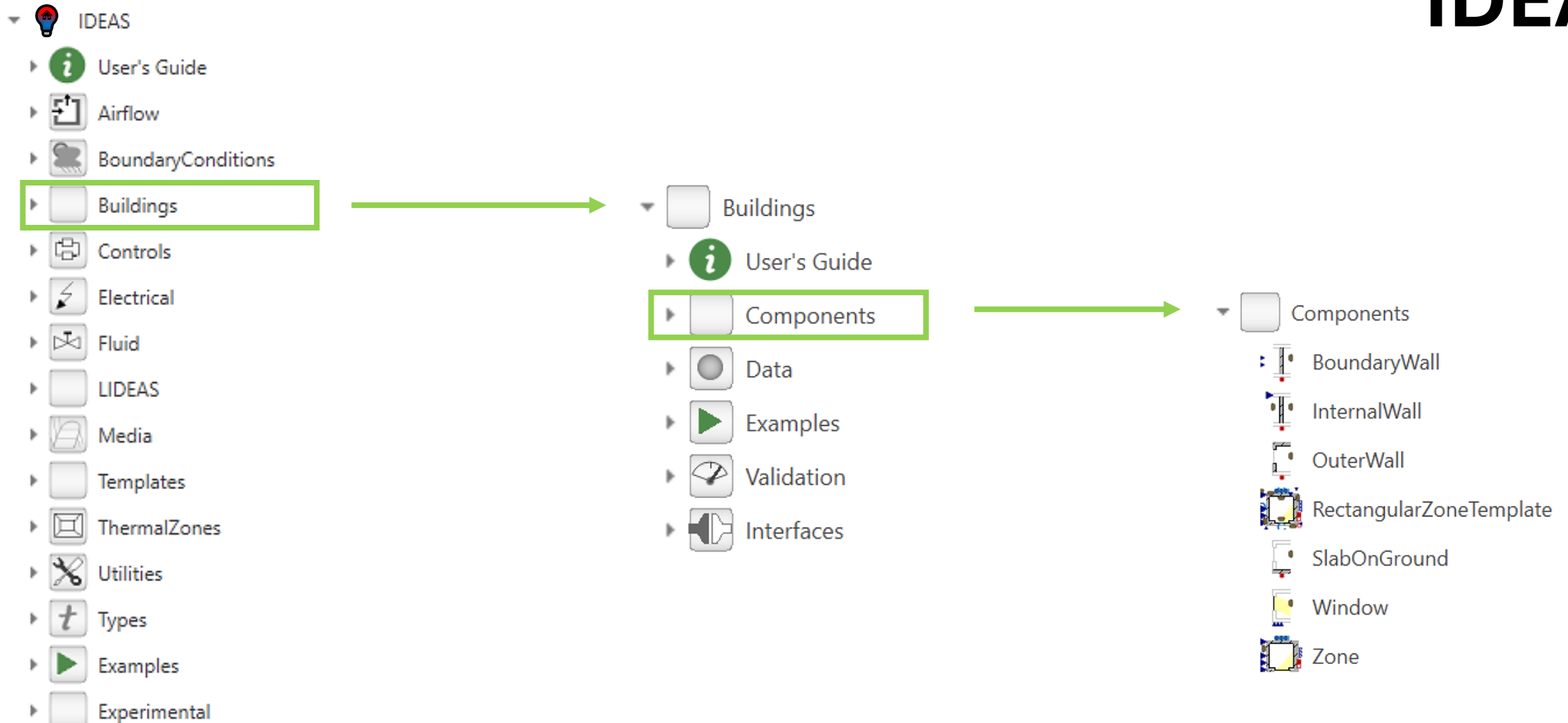
→ https://github.com/open-ideas/_CrashCourse_/blob/master/Exercises/Exercise%202/Latex/Exercise2.pdf



Part 3: IDEAS – Building envelope

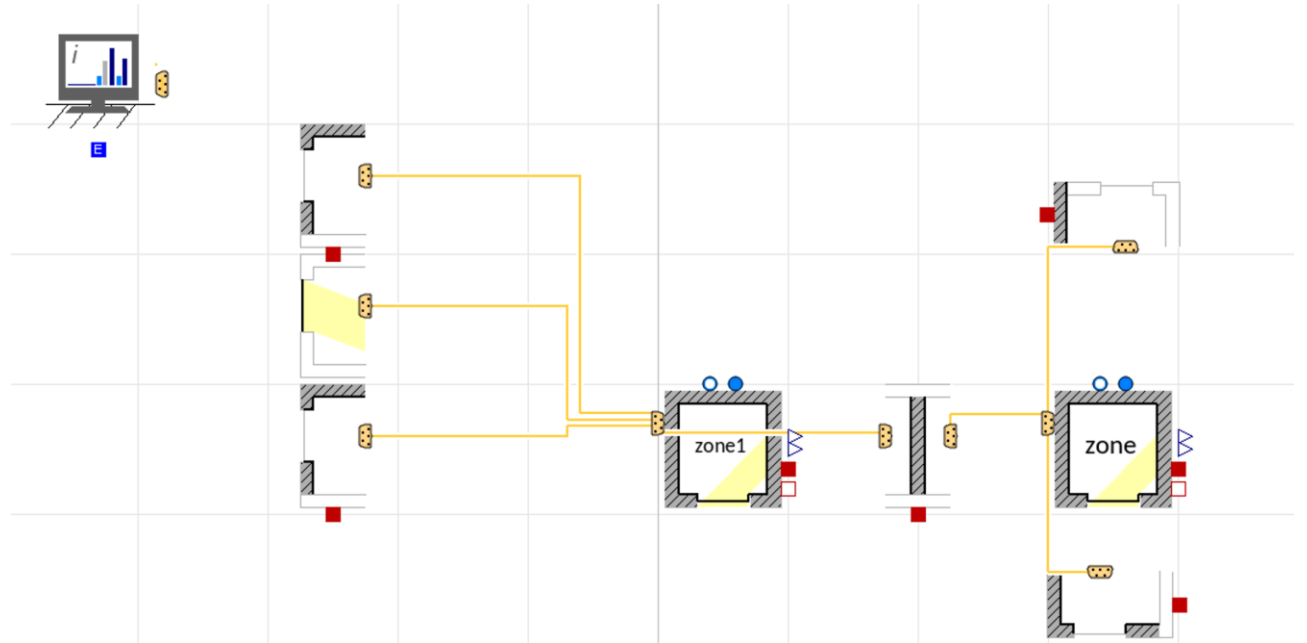
Hannelore Scheipers

IDEAS – Overview

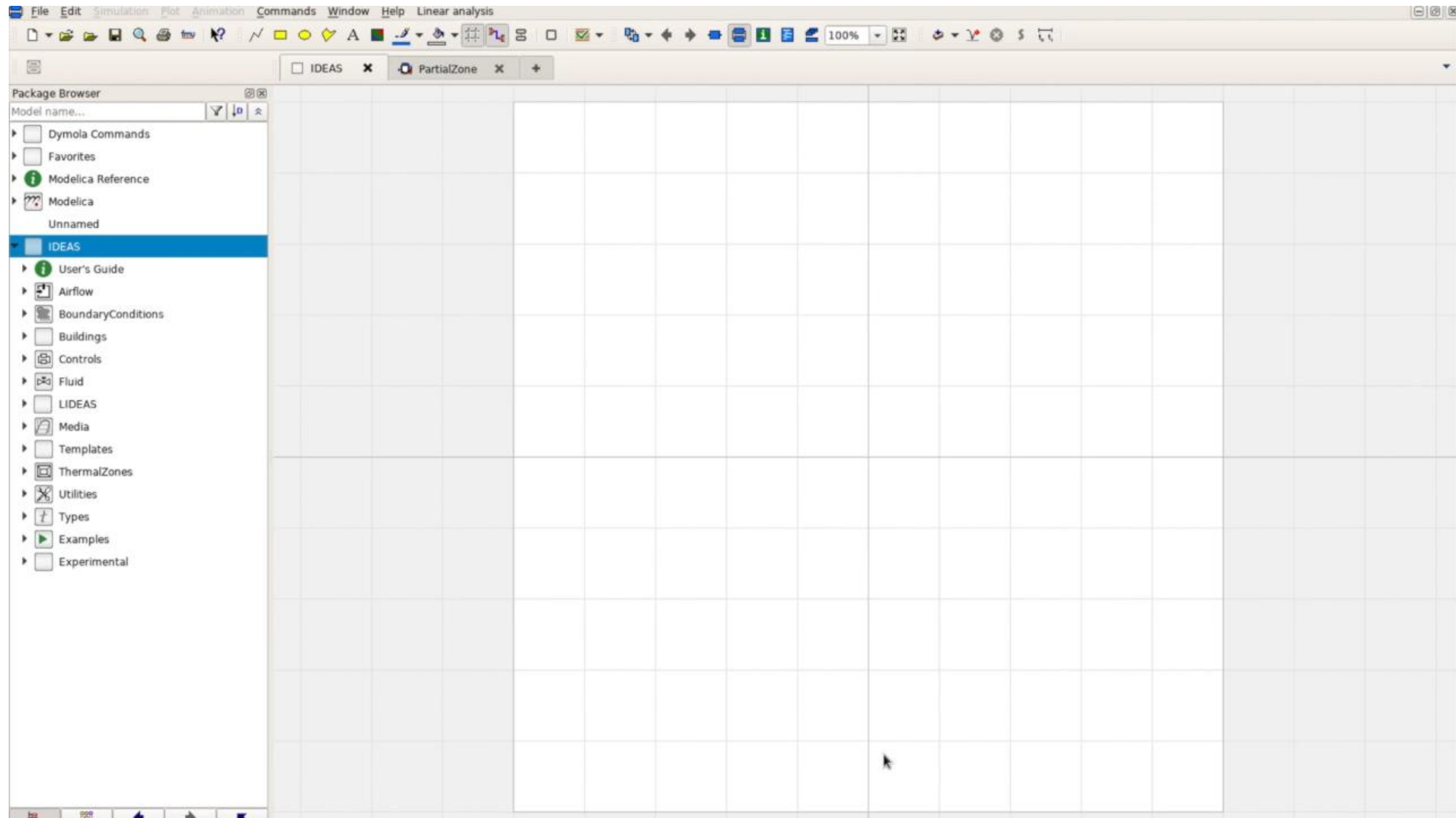


Philosophy

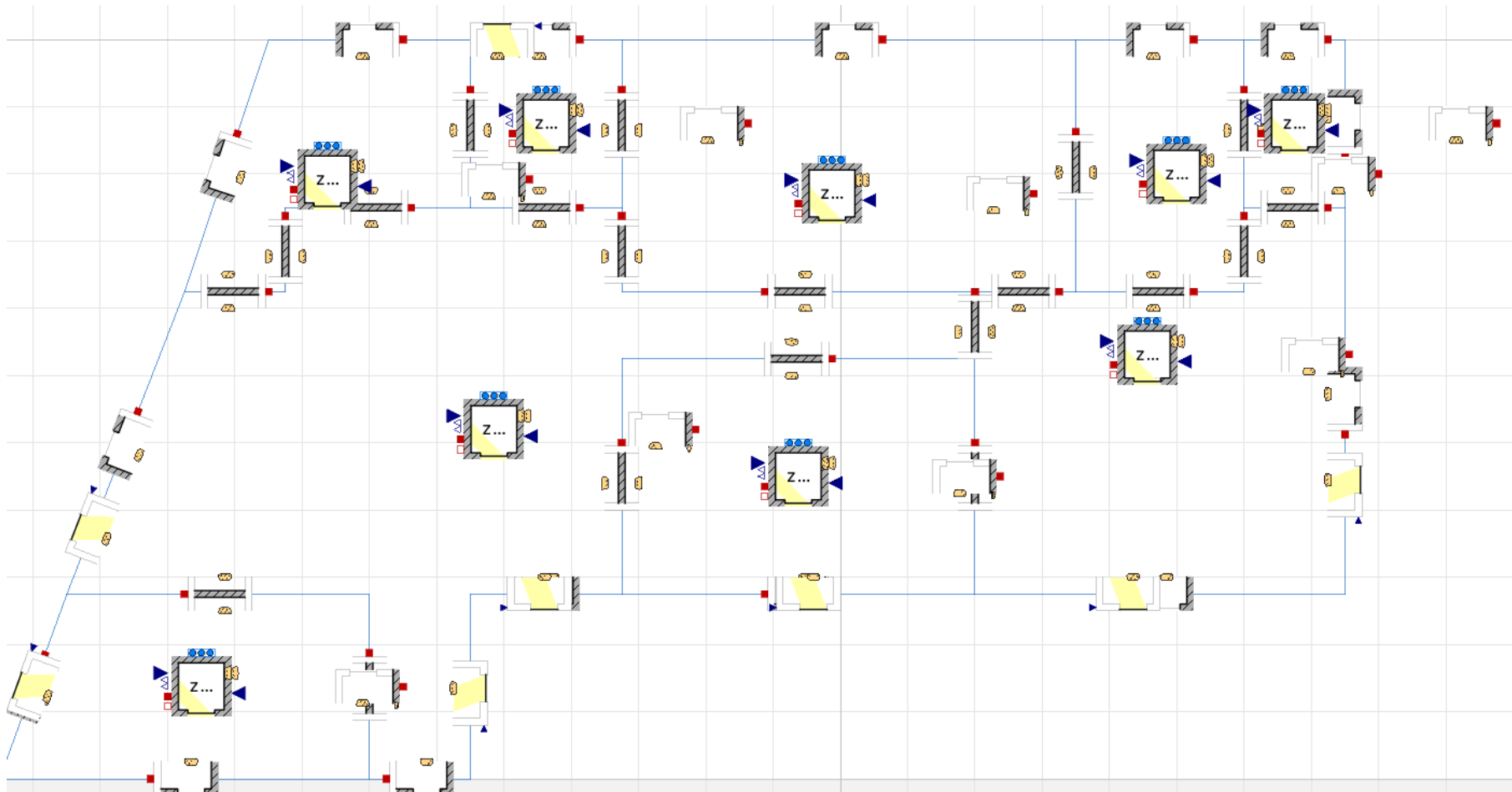
- Direct mapping between physical objects and components (white-box modelling)
- “What you see is what you get”
- Exception: SimInfoManager



IDEAS workflow



Multizone building models



Parameters

Zone

General

Advanced

Airflow

Initialization

Add modifiers

Attributes

Component

Name

zone

Comment

Zone model

Model

Path

IDEAS.Buildings.Components.Zone

Comment

Building zone model

Parameters

Medium

Medium

Medium in the component

nSurf

7

Number of surfaces adjacent to and heat exchanging with the zone

nPorts

2

Number of ports for ventilation connections

energyDynamicsAir

Modelica.Fluid.Types.Dynamics.Fix

Type of energy balance for air model: dynamic (3 initialization options) or steady state

Building physics

V

l*h*w

m³

Total zone air volume

hZone

2.8

m

Zone height: distance between floor and ceiling

hFloor

0

m

Absolute height of zone floor

A

V/hZone

m²

Total conditioned floor area

Occupants (optional)

occNum

redeclare IDEAS.Buildings.Component:

Number of occupants that are present

occTyp

redeclare parameter IDEAS.Buildings.C

Occupancy type, only used for evaluating occupancy model and comfort model

comfort

redeclare IDEAS.Buildings.Component:

Comfort model

Lighting (optional)

rooTyp

redeclare parameter IDEAS.Buildings.C

Room type or function, currently only determines the desired lighting intensity

ligTyp

redeclare parameter IDEAS.Buildings.C

Lighting type, determines the lighting efficacy/efficiency

ligCtr

redeclare IDEAS.Buildings.Components.

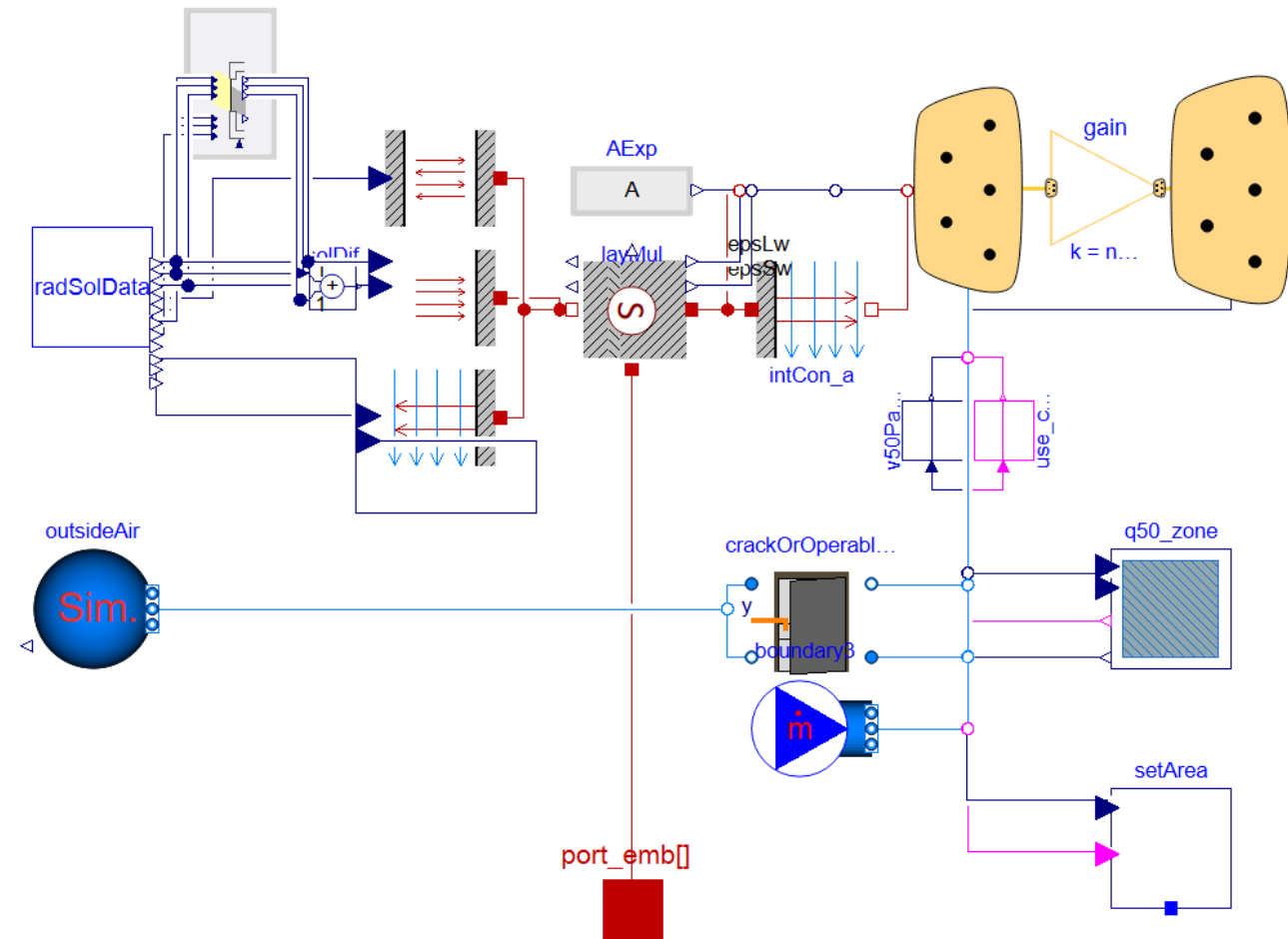
Lighting control type

Wall

General	Advanced	Dynamics	Convection	Airflow	Radiation	Add modifiers	Attributes	
Component Name outerWall Comment Outer wall model Model Path IDEAS.Buildings.Components.OuterWall Comment Opaque building envelope construction								Icon 
Parameters								
incOpt	☐ Wall ☐ Floor ☐ Ceiling ☒ Custom			Tilt angle option from simInfoManager, or custom using inc				
inc	IDEAS.Types.Tilt.Wall			rad	Custom inclination (tilt) angle of the wall, default wall			
aziOpt	☐ South ☐ West ☐ North ☐ East ☒ Custom			Azimuth angle option from simInfoManager, or custom using azi				
azi	IDEAS.Types.Azimuth.W			rad	Custom azimuth angle of the wall, default south			
A	l*h			m ²	Component surface area			
hVertical	if IDEAS.Utilities.Math.Functions.isAngle(in			Vertical surface height, height of the surface projected to the vertical, 0 for floors and ceilings				
hRef_a	if IDEAS.Utilities.Math.Functions.isAngle(in			Height above the zone floor at propsbus_a. Height where the surface starts. e.g. 0 for walls at floor level and floors.				
Construction details								
constructionType	redeclare IDEAS.Buildings.Validation.Data.Constructions.HeavyWall constructionT						Building component material structure	
Building shade								
hasBuildingShade	false			=true, to enable computation of shade cast by opposite building or object				
L	0			m	Distance between object and wall, perpendicular to wall			
dh	0			m	Height difference between top of object and top of wall			
hWal	0			m	Wall height			

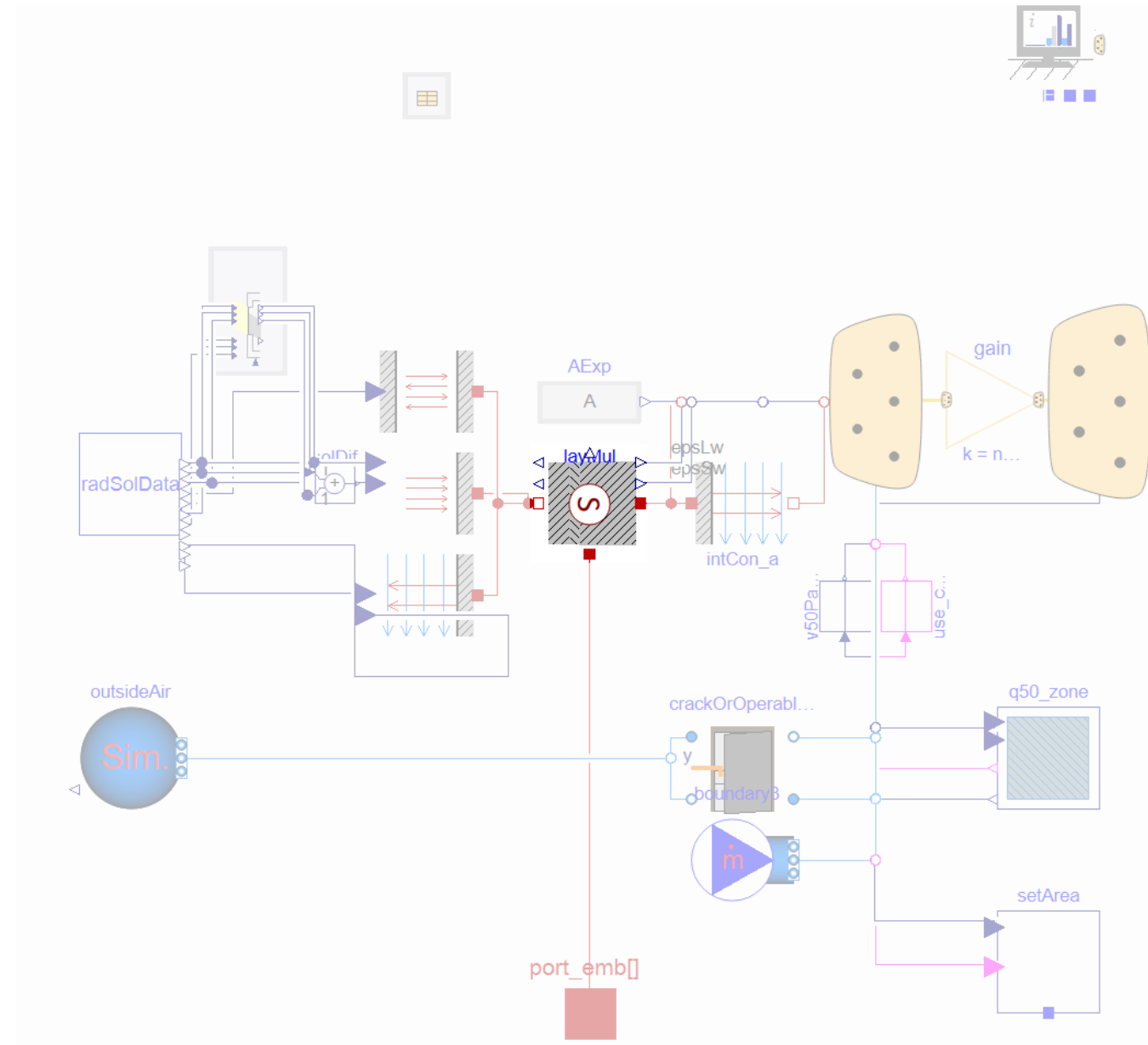
Main building physics

- Conduction, thermal capacity
- Convective heat transfer
- Radiative heat transfer
- Shortwave heat gains (incl. shading)
- Internal heat gains (occupants, lighting)
- **Integrated infiltration and interzonal airflow**



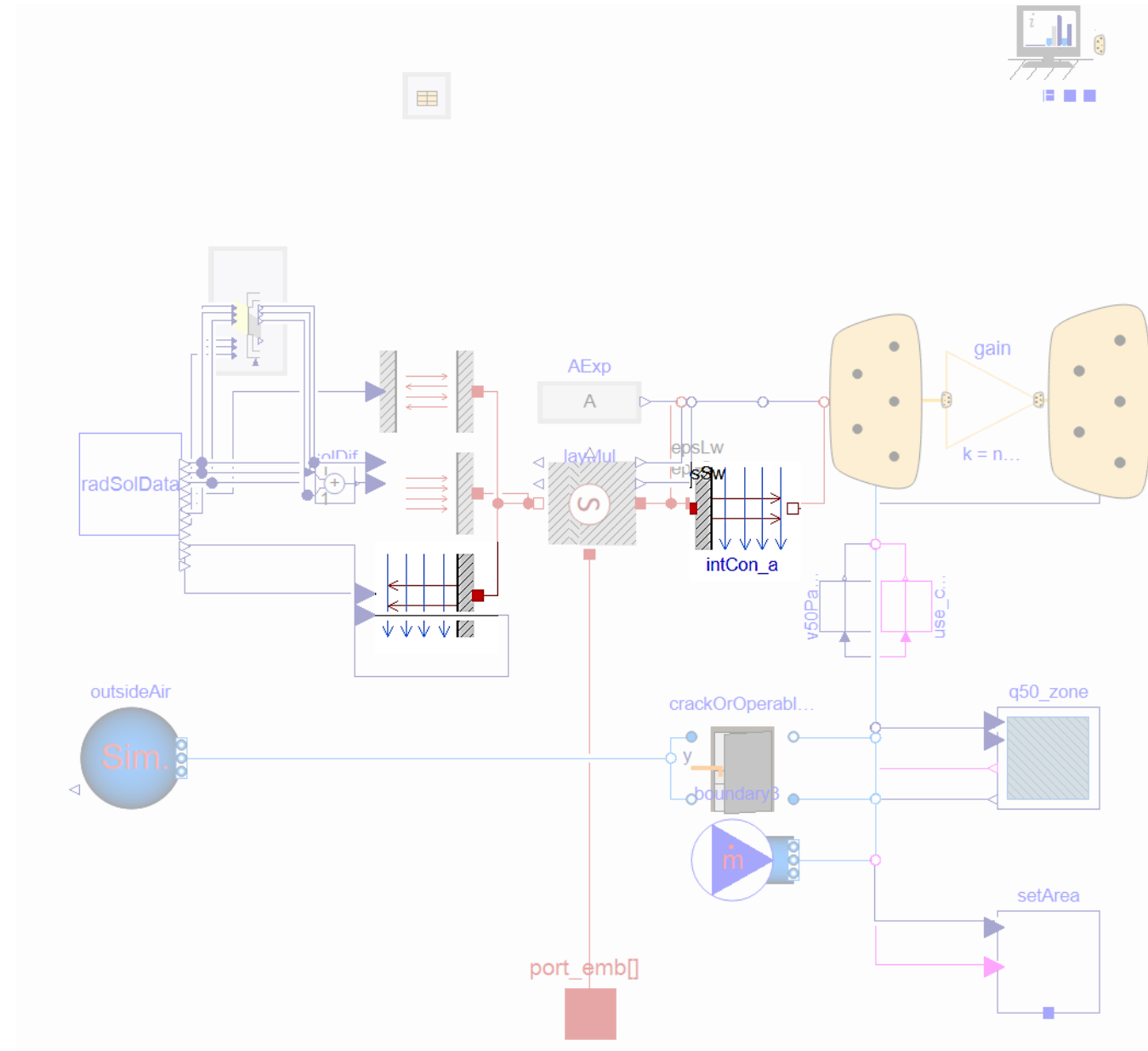
Main building physics

- Conduction, thermal capacity
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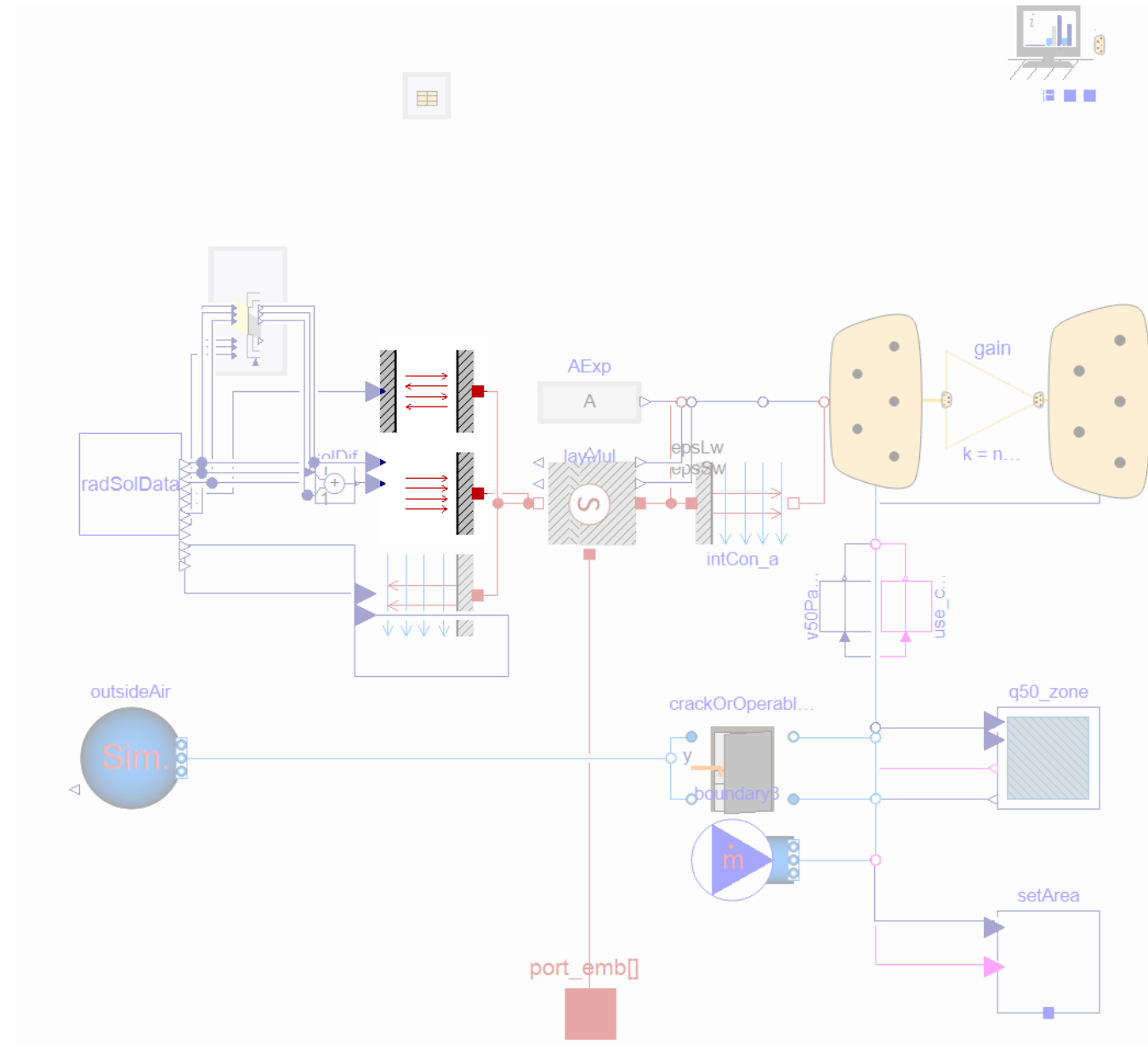
Main building physics

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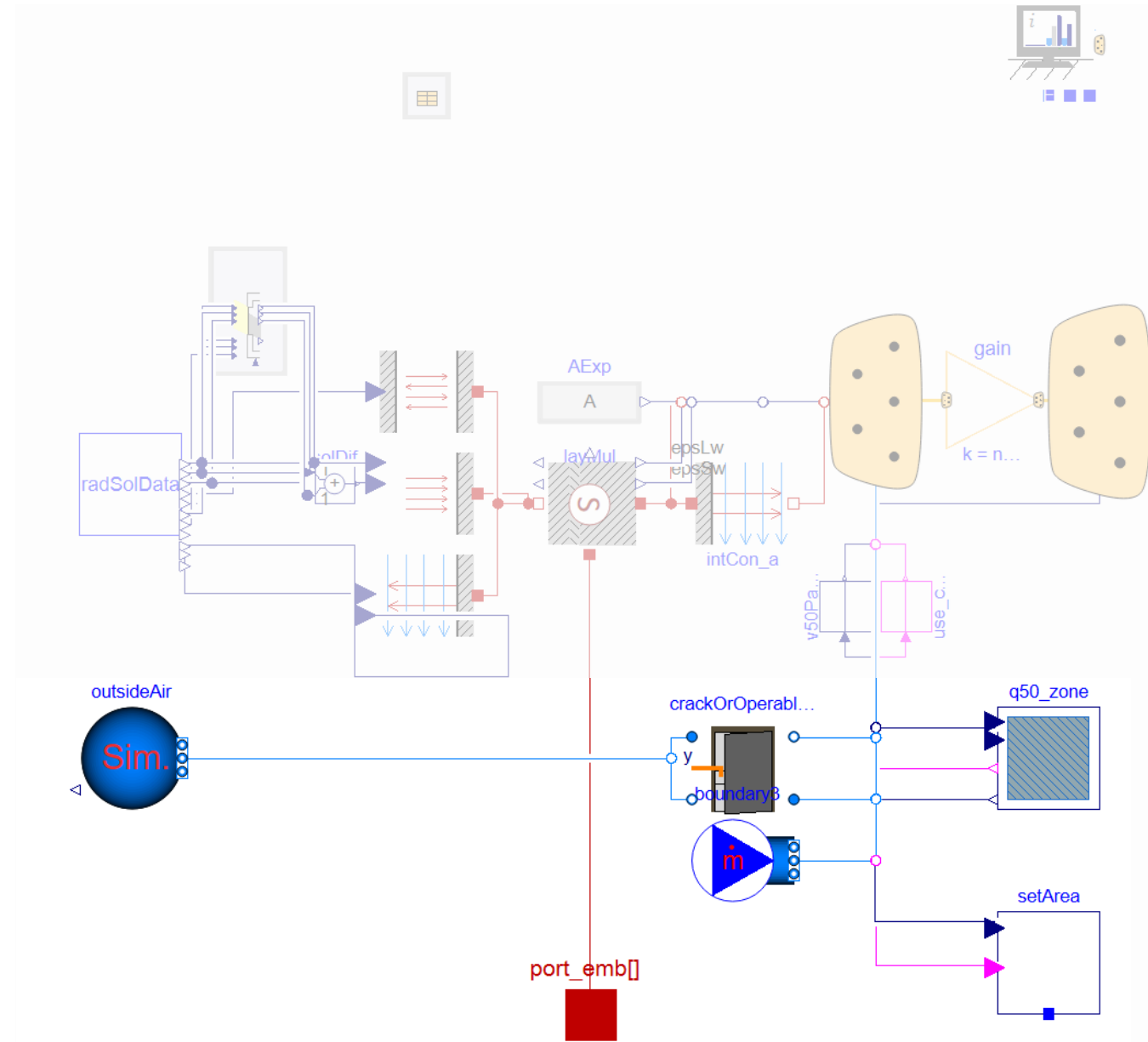
Main building physics

- Conduction, thermal capacity
- Convective heat transfer
- Radiative heat transfer
- Shortwave heat gains (incl. shading)
- Internal heat gains (occupants, lighting)
- Integrated infiltration and interzonal airflow

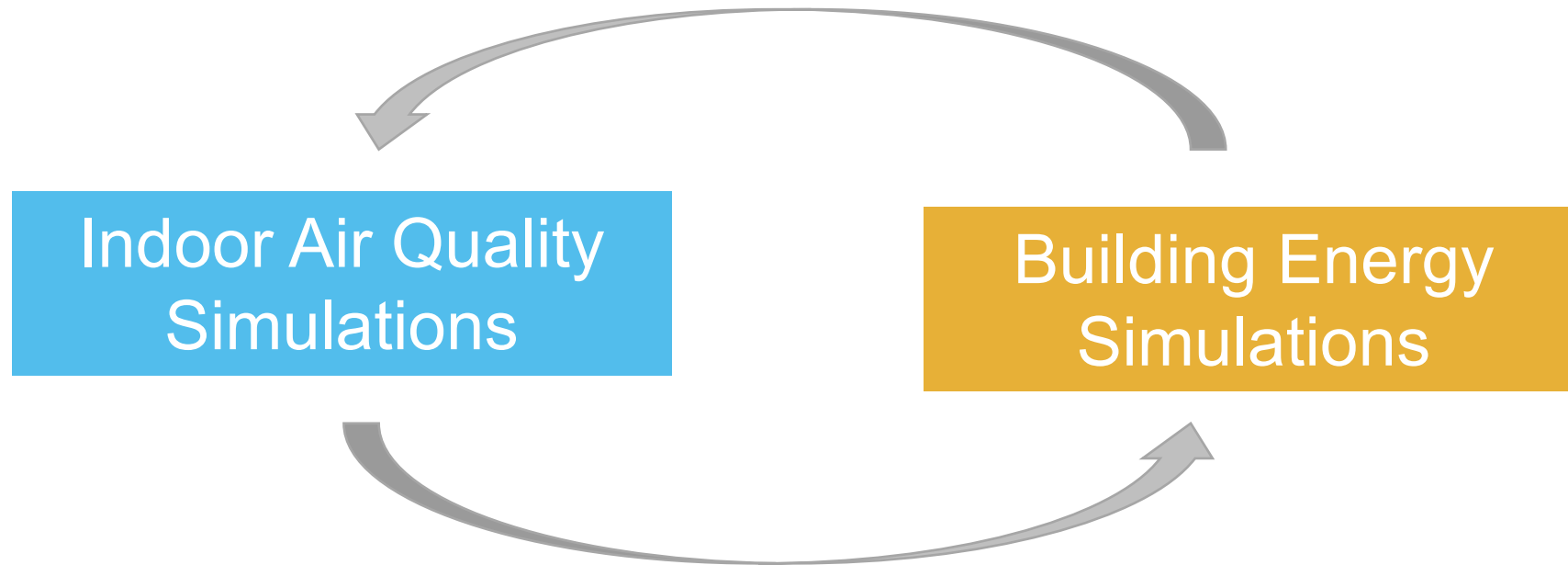


Main building physics

- Conduction, thermal capacity
- Convective heat transfer
- Radiative heat transfer
- Shortwave heat gains (incl. shading)
- Internal heat gains (occupants, lighting)
- **Integrated infiltration and interzonal airflow**

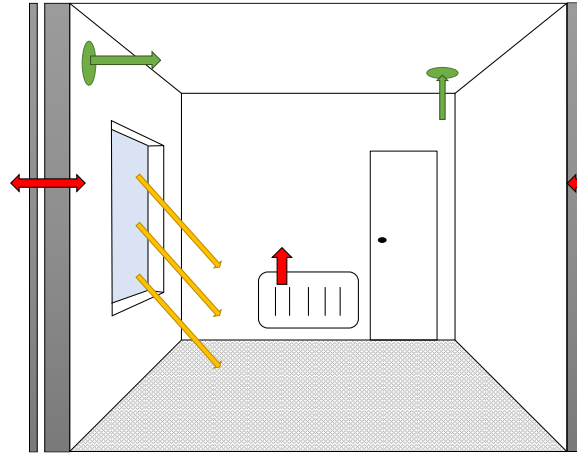


Integrated pressure-driven air flow modelling

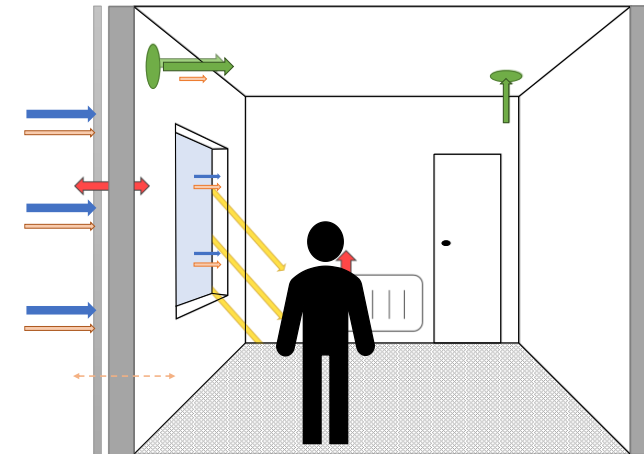
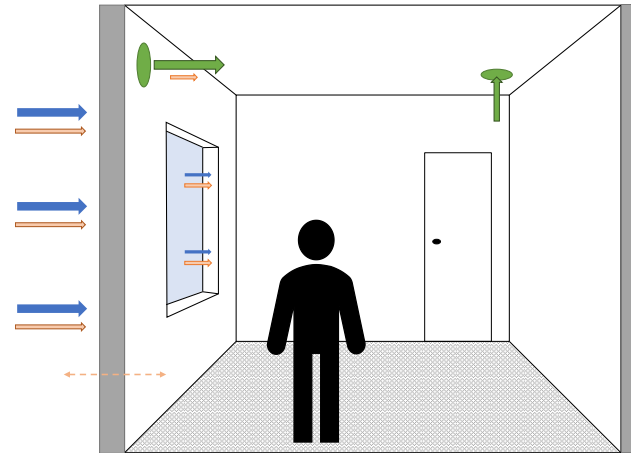


Integrated pressure-driven air flow modelling

Existing building energy
simulation models
[IDEAS]



Airflow elements
[K. De Jonge -> IBPSA]



Integrated coupled energy
and
airflow solution

Exercise 3 – Building envelope model

- See exercise sheet on Github

https://github.com/open-ideas/_CrashCourse_/blob/master/Exercises/Exercise%203/Latex/Exercise3.pdf

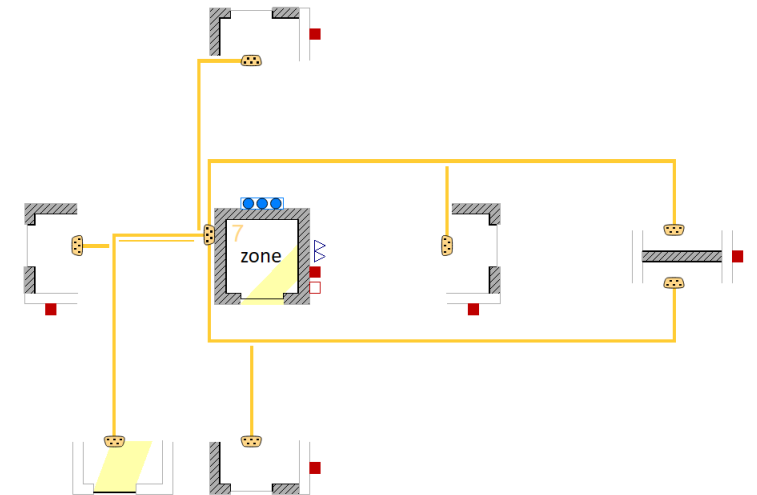
- Make first assignment

1 Building model

Qualitative discussion We will now develop a simple building model that consists of one zone, four walls, a window, a floor and a ceiling. The zone dimensions are 8 *m* (walls with north and south orientation) by 4 *m* and the window is 3 *m* by 1.4 *m*. We use the default zone height (2.8 *m*). We use double glazing and a heavy wall, meaning they have high thermal mass.

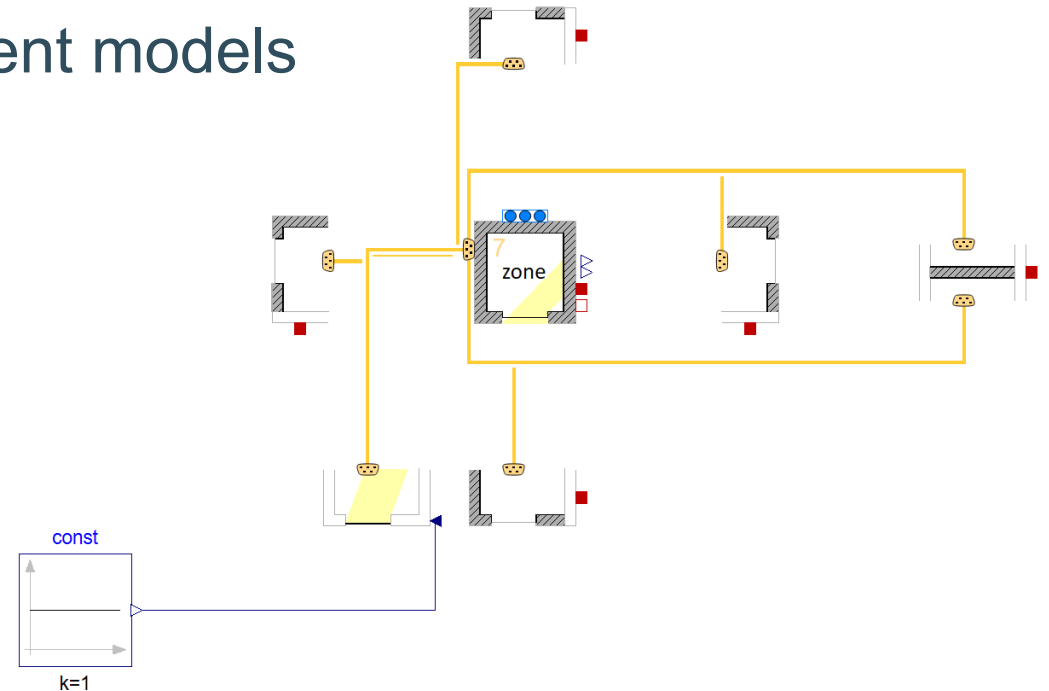
Required models This step requires the main building envelope component models of IDEAS:

- IDEAS.BoundaryConditions.SimInfoManager
- IDEAS.Buildings.Components.Zone
- IDEAS.Buildings.Components.OuterWall
- IDEAS.Buildings.Components.Window
- IDEAS.Buildings.Components.SlabOnGround



Exercise 3 – Building envelope model

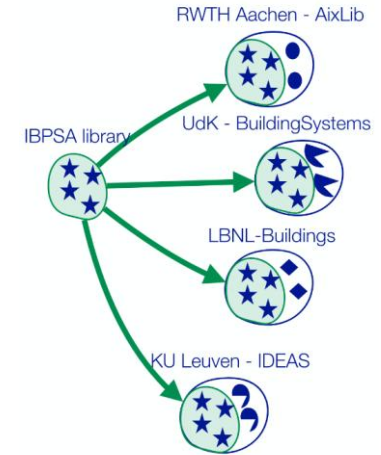
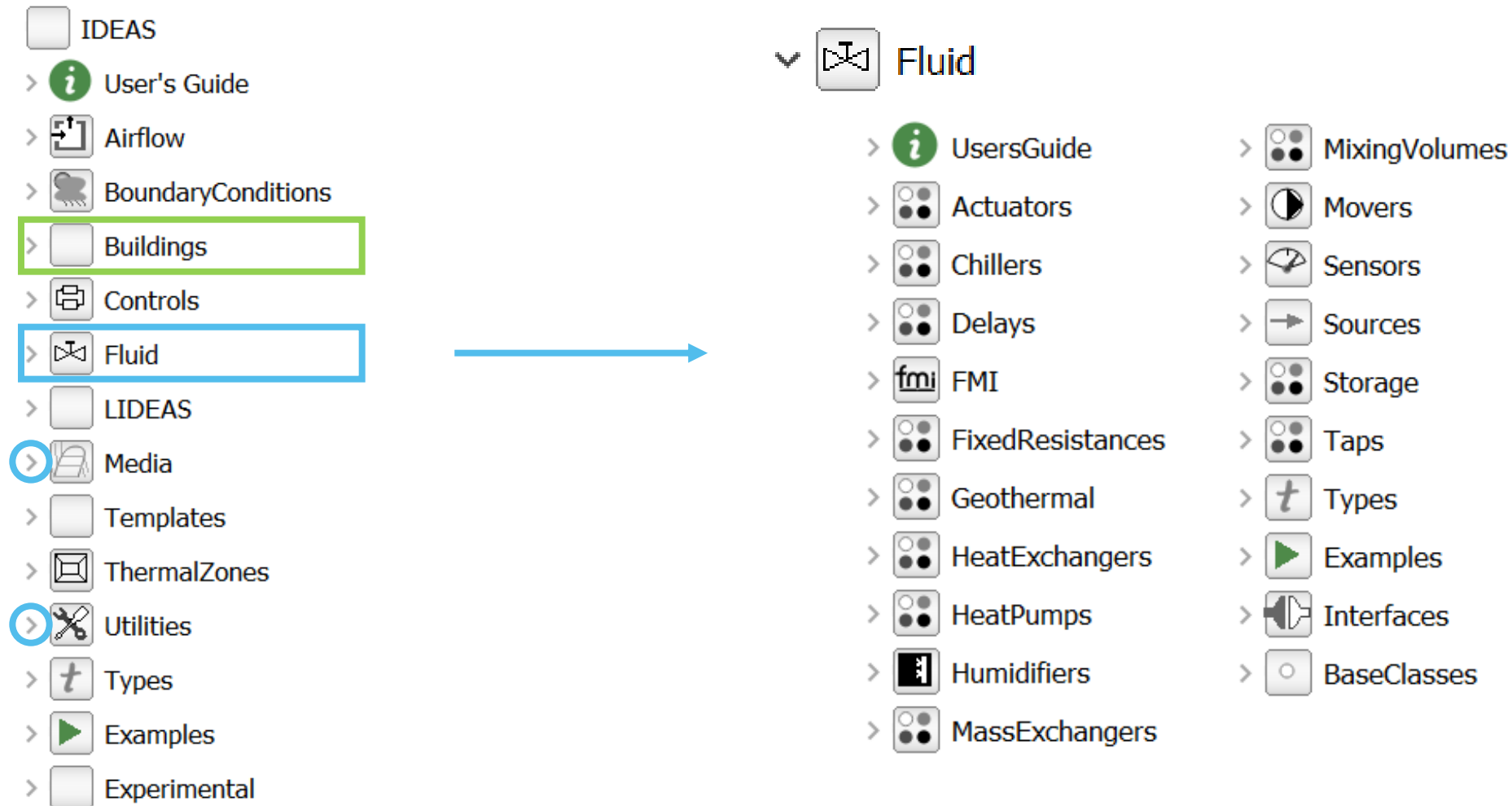
- See exercise sheet on Github
<https://github.com/open-ideas/CrashCourse/blob/master/Exercises/Exercise%203/Latex/Exercise3.pdf>
- Use 'extend' when moving to next assignment
- Use built-in replaceables of IDEAS component models (solar shading, occupancy)



Part 4: IDEAS – HVAC

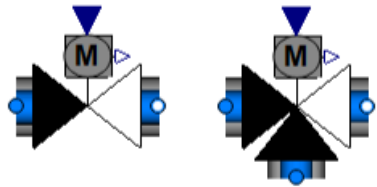
Jaap Neven

IDEAS – HVAC overview

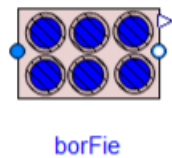


IDEAS – HVAC overview

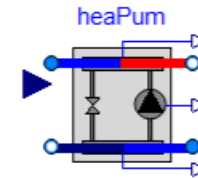
> Actuators



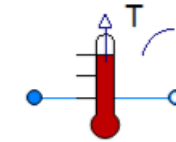
> Geothermal



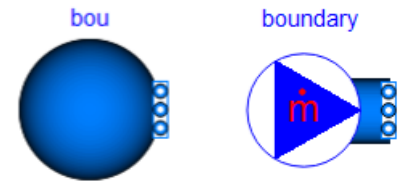
> HeatPumps



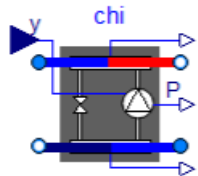
> Sensors



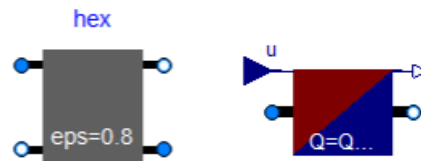
> Sources



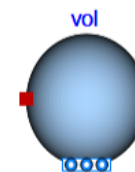
> Chillers



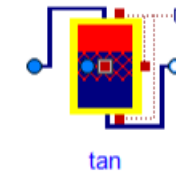
> HeatExchangers



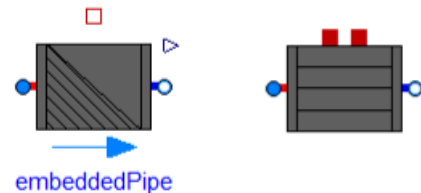
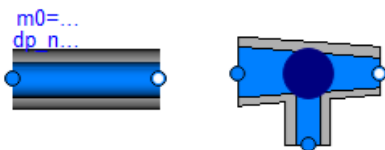
> MixingVolumes



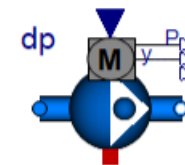
> Storage



> FixedResistances



> Movers

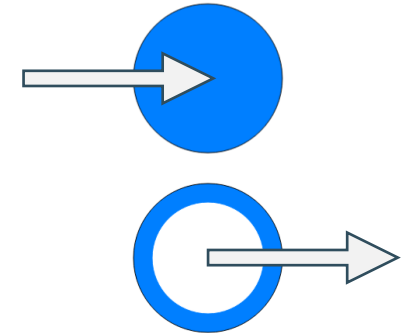


> Taps



Modelica.Fluid.Interfaces.FluidPort

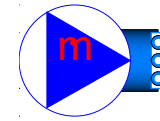
- Potential variable: pressure $\rightarrow$ unique value
- Flow variable: mass flow rate $\rightarrow \Sigma = 0$
- Stream variable: enthalpy $\rightarrow$ characteristic of flow



```
connector FluidPort
  "Interface for quasi one-dimensional fluid flow in a piping network (incompressible or comp.
  replaceable package Medium = Modelica.Media.Interfaces.PartialMedium
    "Medium model" ∃ ;

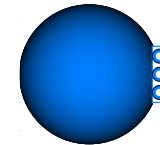
  flow Medium.MassFlowRate m_flow
    "Mass flow rate from the connection point into the component";
  Medium.AbsolutePressure p "Thermodynamic pressure in the connection point";
  stream Medium.SpecificEnthalpy h_outflow
    "Specific thermodynamic enthalpy close to the connection point if m_flow < 0";
  stream Medium.MassFraction Xi_outflow[Medium.nXi]
    "Independent mixture mass fractions m_i/m close to the connection point if m_flow < 0";
  stream Medium.ExtraProperty C_outflow[Medium.nC]
    "Properties c_i/m close to the connection point if m_flow < 0";
end FluidPort;
```

Basic circuit



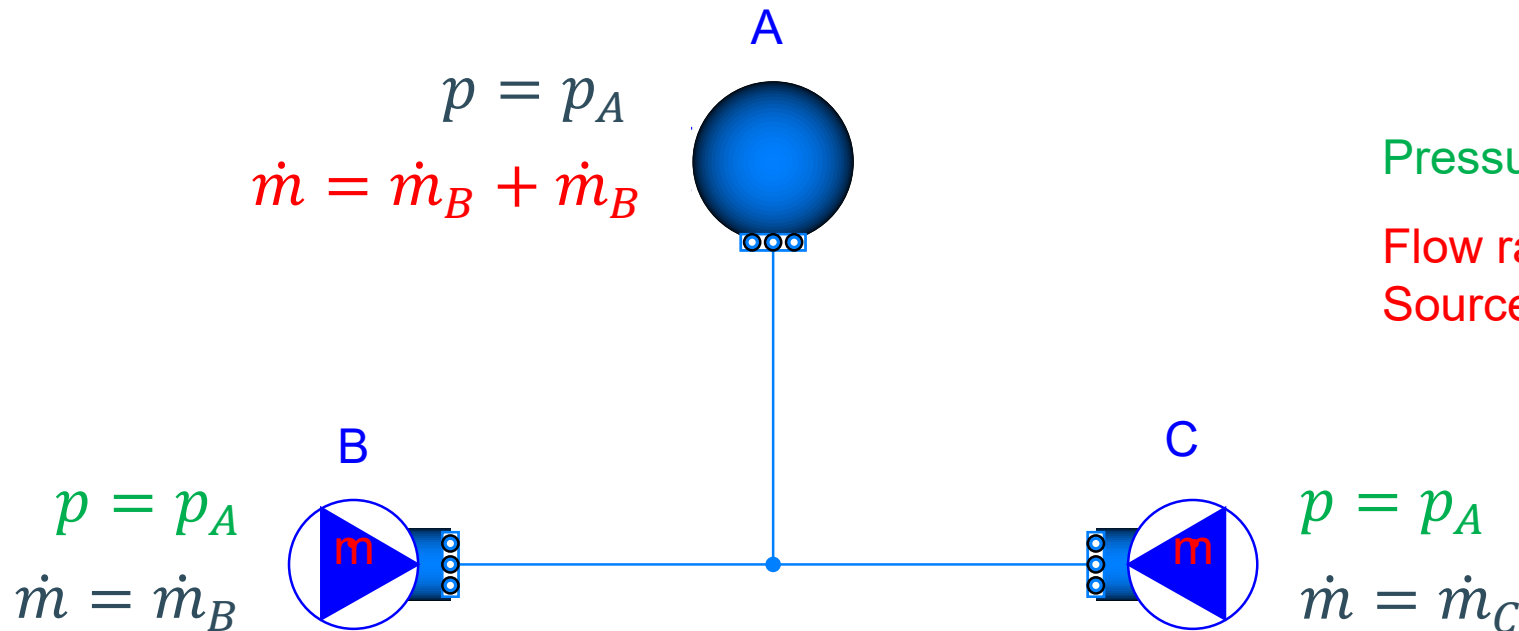
IDEAS.Fluid.Sources.MassFlowSource

→ Sets mass flow rate



IDEAS.Fluid.Sources.Boundary_pT

→ Sets absolute pressure

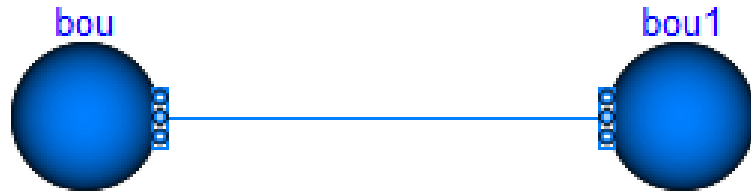


Pressure equals pressure set by Boundary A

Flow rate equals sum of flow rates set by Source B and Source C

Illegal circuits

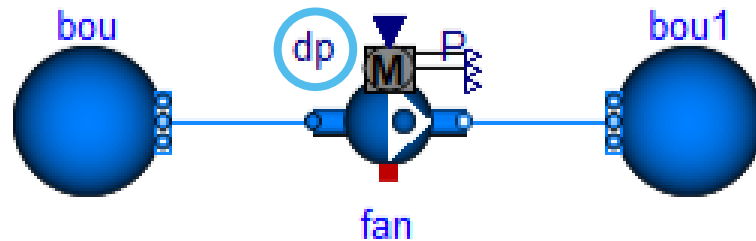
▼ ⚠ The problem is structurally singular



No mass flow rate
& problem when $p_{bou} \neq p_{bou1}$

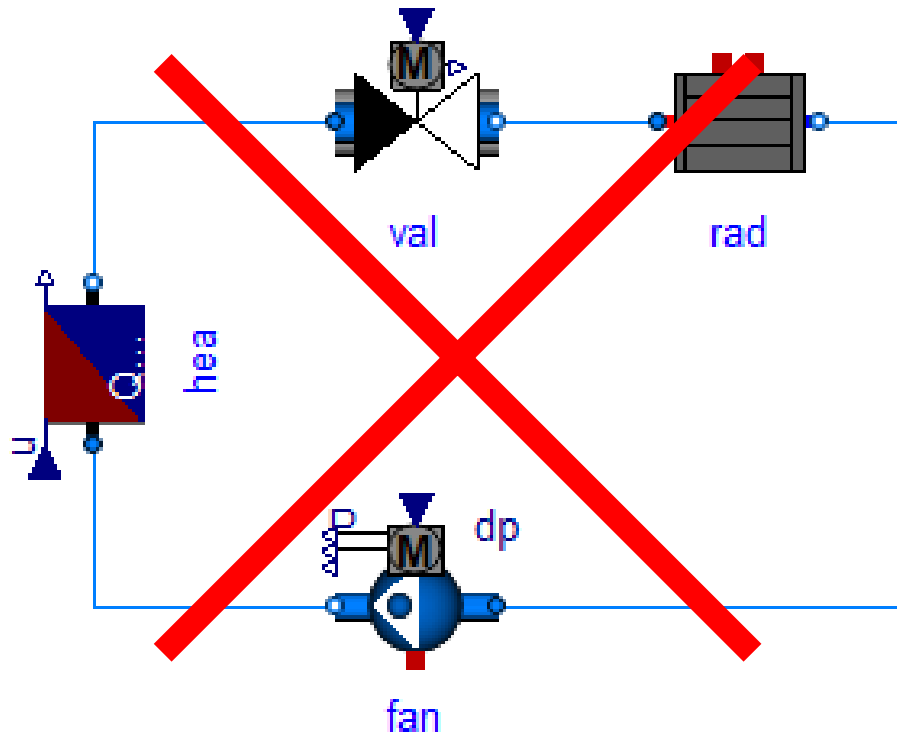


No absolute pressure
& problem when $\dot{m}_{boundary} \neq -\dot{m}_{boundary1}$

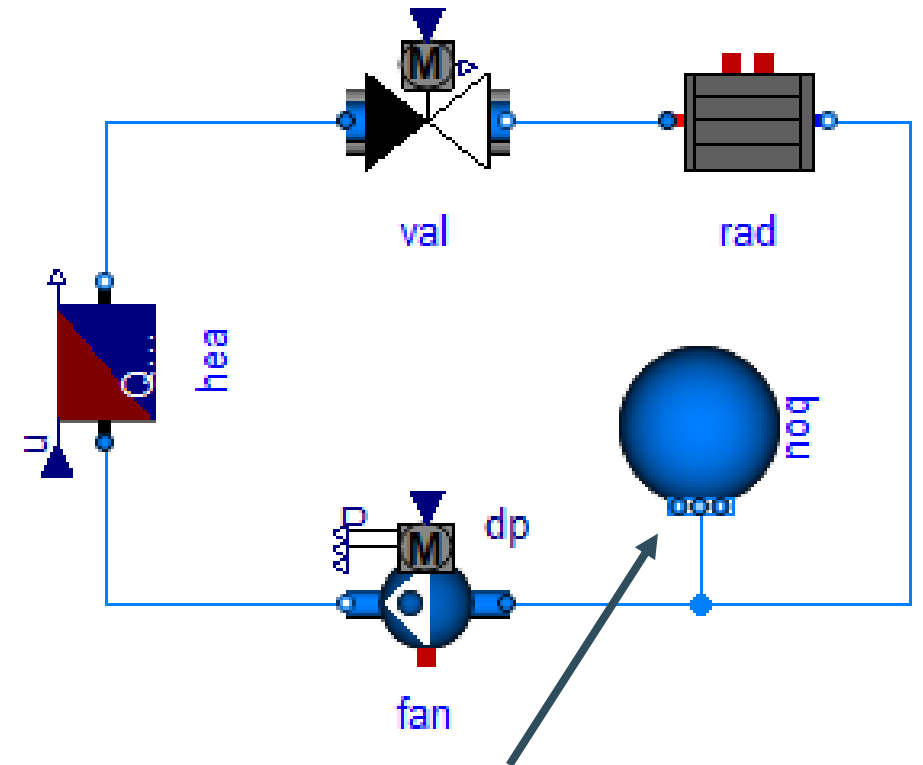
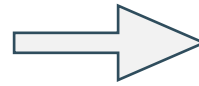


Problem when $p_{bou} + \Delta p_{fan} \neq p_{bou1}$

Simple HVAC circuit



Only pressure drops



Sets absolute pressure

Singularity error

- Modelica is a generic modelling language → errors are general (unclear)
i.e. not tailored to buildings / HVAC
- Important language requirements:
 - # equations = # variables
 - Equations should not contradict each other (e.g. adding both $x = 1$ and $x = 0$)
 - Solution to equations should exist (e.g. don't try $x^2 = -1$)
- These requirements are abstract and are hidden from the user by library developers. However, they pop up sometimes, leading to unclear errors like:

- ▼  The model [IDEAS.Examples.PPD12.Heating](#) is structurally singular.
 - ▼  The problem is structurally singular for the element type Real.
 -  The number of scalar Real unknown elements are 4114.
The number of scalar Real equation elements are 4114.

Singularity error

- ▼  The model [IDEAS.Examples.PPD12.Heating](#) is structurally singular.
- ▼  The problem is structurally singular for the element type Real.
 -  The number of scalar Real unknown elements are 4114.
 - The number of scalar Real equation elements are 4114.

- You don't have to worry about component models when dragging and dropping
- Singularity error occurs, when:
 - Dangling connectors / more than 1 connection
 - # equations $\neq$ # variables
 - Conflicting equations, equations without real solutions
 - Infinite number of solutions (e.g. no absolute pressure set)
- IDEAS.Buildings is fairly robust as long as each zone propsBus connector is connected to exactly one surface propsBus connector.
- IDEAS.Fluid pressure drop circuits can require some experience:
 - Set absolute pressure in flow circuits
 - Don't oversimplify pressure drops

Further reading

- F. Jorissen, G. Reynders, R. Baetens, D. Picard, D. Saelens, and L. Helsen. [Implementation and Verification of the IDEAS Building Energy Simulation Library](#). *Journal of Building Performance Simulation*, **11** (6), 669-688, 2018. doi: 10.1080/19401493.2018.1428361.
- F. Jorissen, M. Wetter, and L. Helsen. Simulation Speed Analysis and Improvements of Modelica Models for Building Energy Simulation. In 11th International Modelica Conference, pages 59–69, Paris, 2015. doi: 10.3384/ecp1511859.
- F. Jorissen, M. Wetter, and L. Helsen. Simplifications for Hydronic System Models in Modelica. *Journal of Building Performance Simulation*, **11** (6), 639-654, 2019.
- F. Jorissen. *Toolchain for Optimal Control and Design of Energy Systems in Buildings*. PhD thesis, Arenberg Doctoral School, KU Leuven, April 2018

Exercise 4 – HVAC model

- See exercise sheet on Github
https://github.com/open-ideas/_CrashCourse_/blob/master/Exercises/Exercise%204/Latex/Exercise4.pdf
- Start from building envelope of step 5 (see *IDEAS.Examples.Tutorial.DetailedHouse*):
 1. Add a geothermal heat pump heating system
 2. Add a heat pump controller
 3. Compute the energy use and export it in a json file
 4. Add a CO₂-controlled ventilation system

